

ABSTRACTED FUNCTIONS FOR BUILDING A TRANSFORMER MODEL

Research & Development Internship Assignment Report

Submitted by: Ashutosh Kumar Singh

Position: Research and Development Intern Applicant

Company: NVSKZEN

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1. Introduction to Neural Networks

Artificial Neural Networks (ANNs) are computational models inspired by the structure and functioning of the human brain. These networks are designed to recognize patterns, process information, and make predictions by learning from data.

A neural network consists of:

- Input Layer
- Hidden Layers
- Output Layer

Each neuron receives input values, applies weights, adds bias, and passes the result through an activation function.

Basic Working of a Neural Network

The mathematical representation of a neuron is:

$$Y = f(WX + b)$$

Where:

- X = Input vector
- W = Weight matrix
- b = Bias
- f = Activation function
- Y = Output

Neural networks learn by minimizing the error between predicted and actual outputs using optimization techniques such as Gradient Descent and Backpropagation.

2. Introduction to Transformer Models

Transformer models are one of the most important breakthroughs in Deep Learning and Natural Language Processing (NLP). Introduced in the research paper:

"Attention Is All You Need" (Vaswani et al., 2017)

Transformers replaced recurrent architectures such as:

- Recurrent Neural Networks (RNNs)
- Long Short-Term Memory Networks (LSTMs)
- Gated Recurrent Units (GRUs)

The Transformer architecture is based entirely on the concept of **Attention Mechanisms**, enabling parallel processing and significantly improving training efficiency.

Transformers are widely used in:

- ChatGPT
- BERT
- GPT Series
- T5
- Vision Transformers
- Machine Translation
- Text Summarization
- Speech Recognition
- Recommendation Systems

3. Evolution from RNNs to Transformers

Problems with RNNs

Traditional sequential models process tokens one-by-one.

Limitations include:

1. Vanishing Gradient Problem
2. Difficulty handling long-term dependencies
3. Slow training due to sequential computation
4. Poor parallelization

How Transformers Solve These Problems

Transformers:

- Process all words simultaneously
- Capture long-range dependencies efficiently
- Use attention instead of recurrence
- Train significantly faster on GPUs

This led to major improvements in modern AI systems.

4. Concept of Abstracted Functions

Definition

Abstracted functions are reusable modular components designed to perform a specific task independently.

Instead of writing the entire Transformer logic in one large block of code, we divide the implementation into smaller functions.

Examples:

- `create_attention_mask()`
- `positional_encoding()`
- `multi_head_attention()`
- `encoder_layer()`
- `decoder_layer()`

Why Abstraction is Important

Abstraction improves:

1. Readability

The code becomes easier to understand.

2. Reusability

Functions can be reused across multiple projects.

3. Scalability

Large architectures become manageable.

4. Debugging

Errors can be isolated easily.

5. Collaboration

Teams can work on independent modules.

6. Maintainability

Future updates become simpler.

In research and production systems, abstraction is essential for building scalable AI pipelines.

5. Importance of Abstraction in Deep Learning

Deep learning architectures can become extremely complex.

Without abstraction:

- Code becomes difficult to debug
- Maintenance becomes challenging
- Reusability decreases
- Research experimentation slows down

With abstraction:

- Individual layers can be tested independently
- Hyperparameters can be modified easily
- Models can be extended efficiently
- Research reproducibility improves

Modern frameworks such as:

- PyTorch
- TensorFlow
- Keras

are themselves built using abstraction principles.

6. Transformer Architecture Overview

A Transformer primarily consists of:

1. Encoder Stack
2. Decoder Stack
3. Multi-Head Attention
4. Positional Encoding
5. Feed Forward Networks
6. Residual Connections
7. Layer Normalization

High-Level Flow

Input Sentence → Embedding → Positional Encoding → Encoder → Decoder → Output Probabilities

7. Components of the Transformer

7.1 Input Embedding

Neural networks cannot process raw text directly.

Words are converted into dense vectors called embeddings.

Example:

Word	Vector Representation
AI	[0.12, 0.44, 0.91]
Deep	[0.81, 0.22, 0.17]

Embeddings capture semantic meaning.

Embedding Formula

$$Embedding = Lookup(Token\ ID)$$

7.2 Positional Encoding

Transformers process all tokens simultaneously.

Therefore, positional information must be added manually.

Formula

For even indices:

$$PE(pos, 2i) = \sin(pos/10000^{2i/d_{model}})$$

For odd indices:

$$PE(pos, 2i + 1) = \cos(pos/10000^{2i/d_{model}})$$

This helps the model understand word order.

7.3 Self-Attention Mechanism

Self-attention enables each token to focus on relevant words.

Example:

Sentence:

"The animal didn't cross the road because it was tired."

The word "it" attends to "animal".

Query, Key, and Value

Each token generates:

- Query (Q)
- Key (K)
- Value (V)

Attention Formula

$$Attention(Q, K, V) = softmax\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

Where:

- Q = Query matrix
- K = Key matrix
- V = Value matrix
- d_k = Dimension scaling factor

This mechanism allows contextual understanding.

7.4 Multi-Head Attention

Instead of using one attention mechanism, Transformers use multiple attention heads.

Each head learns different relationships.

For example:

- One head learns grammar

- Another learns semantics
- Another learns positional relations

Formula

$$MultiHead(Q, K, V) = Concat(head_1, \dots, head_h)W^O$$

This improves representation learning.

7.5 Feed Forward Network

After attention, data passes through a fully connected neural network.

Formula

$$FFN(x) = \max(0, xW_1 + b_1)W_2 + b_2$$

This introduces non-linearity and deeper feature extraction.

7.6 Residual Connections and Layer Normalization

Residual connections help avoid vanishing gradients.

Formula

$$Output = LayerNorm(x + Sublayer(x))$$

Benefits:

- Faster convergence
 - Stable gradients
 - Improved training
-

7.7 Encoder

The encoder processes input sequences.

Each encoder layer contains:

1. Multi-head self-attention
2. Feed forward network

3. Residual connections
4. Layer normalization

Multiple encoder layers are stacked.

7.8 Decoder

The decoder generates output tokens.

It contains:

1. Masked multi-head attention
2. Encoder-decoder attention
3. Feed forward network
4. Residual connections

The decoder predicts one token at a time.

8. Mathematical Foundations

The Transformer architecture is primarily built upon matrix operations, probability distributions, and optimization techniques. Among all mathematical concepts used in the Transformer, the scaled dot-product attention mechanism is the most important.

Attention Formula

The scaled dot-product attention mechanism is mathematically represented as:

$$Attention(Q, K, V) = softmax \left(\frac{QK^T}{\sqrt{d_k}} \right) V$$

Where:

- Q = Query Matrix
- K = Key Matrix
- V = Value Matrix
- d_k = Dimension of the key vectors used for scaling

The attention mechanism enables the model to determine which words in a sequence are more important while processing contextual information.

This capability allows Transformers to capture long-range dependencies and semantic relationships effectively.

9. Model Design and Architecture

Objective

The objective of this project is to build a simplified Transformer model using modular abstracted functions.

Design Goals

1. Simplicity
2. Modularity
3. Reusability
4. Scalability
5. Research-oriented structure

Architecture Pipeline

1. Tokenization
2. Embedding Layer
3. Positional Encoding
4. Multi-Head Attention
5. Encoder Layers
6. Decoder Layers
7. Output Linear Layer
8. Softmax Prediction

10. Implementation Using Abstracted Functions

The project was divided into multiple reusable functions.

Core Functions

Function Name	Purpose
<code>positional_encoding()</code>	Adds positional information
<code>scaled_dot_product_attention()</code>	Computes attention scores
<code>multi_head_attention()</code>	Applies multiple attention heads
<code>feed_forward_network()</code>	Fully connected layers
<code>encoder_layer()</code>	Single encoder block
<code>decoder_layer()</code>	Single decoder block

Function Name	Purpose
transformer()	Complete Transformer model
train_step()	Performs training

This modular design makes experimentation easier.

11. Complete Python Implementation

Libraries Used

```
import torch
import torch.nn as nn
import torch.optim as optim
import math
```

Positional Encoding

```
class PositionalEncoding(nn.Module):
    def __init__(self, d_model, max_len=5000):
        super(PositionalEncoding, self).__init__()

        pe = torch.zeros(max_len, d_model)
        position = torch.arange(0, max_len).unsqueeze(1)

        div_term = torch.exp(
            torch.arange(0, d_model, 2) *
            (-math.log(10000.0) / d_model)
        )

        pe[:, 0::2] = torch.sin(position * div_term)
        pe[:, 1::2] = torch.cos(position * div_term)

        pe = pe.unsqueeze(0)
        self.register_buffer('pe', pe)

    def forward(self, x):
        return x + self.pe[:, :x.size(1)]
```

Scaled Dot Product Attention

```
class ScaledDotProductAttention(nn.Module):
    def __init__(self):
        super().__init__()

    def forward(self, Q, K, V, mask=None):
        d_k = Q.size(-1)

        scores = torch.matmul(Q, K.transpose(-2, -1)) / math.sqrt(d_k)

        if mask is not None:
            scores = scores.masked_fill(mask == 0, -1e9)

        attention = torch.softmax(scores, dim=-1)

        output = torch.matmul(attention, V)

        return output, attention
```

Multi-Head Attention

```
class MultiHeadAttention(nn.Module):
    def __init__(self, d_model, num_heads):
        super().__init__()

        self.num_heads = num_heads
        self.d_model = d_model
        self.d_k = d_model // num_heads

        self.W_q = nn.Linear(d_model, d_model)
        self.W_k = nn.Linear(d_model, d_model)
        self.W_v = nn.Linear(d_model, d_model)

        self.fc = nn.Linear(d_model, d_model)

        self.attention = ScaledDotProductAttention()

    def split_heads(self, x, batch_size):
        x = x.view(batch_size, -1, self.num_heads, self.d_k)
        return x.transpose(1, 2)

    def forward(self, Q, K, V, mask=None):
```

```

batch_size = Q.size(0)

Q = self.split_heads(self.W_q(Q), batch_size)
K = self.split_heads(self.W_k(K), batch_size)
V = self.split_heads(self.W_v(V), batch_size)

attention_output, attention = self.attention(Q, K, V, mask)

attention_output = attention_output.transpose(1, 2).contiguous()

attention_output = attention_output.view(
    batch_size, -1, self.d_model
)

output = self.fc(attention_output)

return output

```

Feed Forward Network

```

class FeedForward(nn.Module):
    def __init__(self, d_model, d_ff):
        super().__init__()

        self.linear1 = nn.Linear(d_model, d_ff)
        self.linear2 = nn.Linear(d_ff, d_model)
        self.relu = nn.ReLU()

    def forward(self, x):
        return self.linear2(self.relu(self.linear1(x)))

```

Encoder Layer

```

class EncoderLayer(nn.Module):
    def __init__(self, d_model, num_heads, d_ff, dropout=0.1):
        super().__init__()

        self.mha = MultiHeadAttention(d_model, num_heads)
        self.ffn = FeedForward(d_model, d_ff)

        self.norm1 = nn.LayerNorm(d_model)

```

```

        self.norm2 = nn.LayerNorm(d_model)

        self.dropout = nn.Dropout(dropout)

    def forward(self, x, mask=None):
        attn_output = self.mha(x, x, x, mask)

        x = self.norm1(x + self.dropout(attn_output))

        ffn_output = self.ffn(x)

        x = self.norm2(x + self.dropout(ffn_output))

    return x

```

Transformer Model

```

class Transformer(nn.Module):
    def __init__(self,
                  vocab_size,
                  d_model,
                  num_heads,
                  num_layers,
                  d_ff,
                  max_len):

        super().__init__()

        self.embedding = nn.Embedding(vocab_size, d_model)

        self.pos_encoding = PositionalEncoding(d_model, max_len)

        self.layers = nn.ModuleList([
            EncoderLayer(d_model, num_heads, d_ff)
            for _ in range(num_layers)
        ])

        self.fc_out = nn.Linear(d_model, vocab_size)

    def forward(self, x):
        x = self.embedding(x)

        x = self.pos_encoding(x)

```

```
    for layer in self.layers:
        x = layer(x)

    output = self.fc_out(x)

    return output
```

Training Loop

```
model = Transformer(
    vocab_size=10000,
    d_model=128,
    num_heads=8,
    num_layers=4,
    d_ff=512,
    max_len=100
)

criterion = nn.CrossEntropyLoss()
optimizer = optim.Adam(model.parameters(), lr=0.001)

for epoch in range(10):
    optimizer.zero_grad()

    output = model(input_data)

    loss = criterion(
        output.view(-1, 10000),
        target.view(-1)
    )

    loss.backward()

    optimizer.step()

    print(f"Epoch {epoch+1}, Loss: {loss.item()}")
```

12. Training Process

Step 1: Data Collection

Text datasets are collected.

Possible datasets:

- Wikipedia text
- OpenSubtitles
- IMDB Reviews
- Custom text corpus

Step 2: Tokenization

Sentences are converted into tokens.

Example:

"AI is powerful"

→ ["AI", "is", "powerful"]

Step 3: Vocabulary Building

Each token receives a unique integer ID.

Step 4: Embedding Conversion

Token IDs are transformed into dense vectors.

Step 5: Forward Propagation

Data flows through the Transformer.

Step 6: Loss Computation

Cross entropy loss measures prediction error.

Step 7: Backpropagation

Gradients are computed.

Step 8: Optimization

Weights are updated using Adam optimizer.

Step 9: Epoch Repetition

Training repeats until convergence.

13. Dataset Preparation

Data Preprocessing Steps

1. Lowercasing
2. Removing punctuation
3. Removing stopwords
4. Tokenization
5. Padding sequences
6. Creating attention masks

Importance of Preprocessing

Proper preprocessing:

- Reduces noise
 - Improves learning
 - Increases model accuracy
 - Stabilizes training
-

14. Hyperparameter Selection

Hyperparameter	Value
Embedding Dimension	128
Number of Heads	8
Encoder Layers	4
Feed Forward Size	512
Batch Size	32
Learning Rate	0.001

Hyperparameter	Value
Optimizer	Adam
Epochs	10

Why Hyperparameters Matter

Hyperparameters strongly influence:

- Training speed
 - Model stability
 - Accuracy
 - Generalization
-

15. Loss Function and Optimization

Cross Entropy Loss

Used for classification-based token prediction.

Advantages:

- Stable training
- Effective probability learning
- Suitable for NLP tasks

Adam Optimizer

Adam combines:

- Momentum
- Adaptive learning rates

Advantages:

- Faster convergence
 - Better stability
 - Widely used in Transformer training
-

16. Challenges Faced and Solutions

Challenge 1: Understanding Attention Mechanism

Problem

Attention calculations involve matrix multiplications and scaling.

Solution

Studied research papers and visualized tensor dimensions step-by-step.

Challenge 2: Tensor Shape Mismatch

Problem

Errors occurred during multi-head splitting.

Solution

Used debugging prints and verified tensor dimensions carefully.

Challenge 3: High Computational Cost

Problem

Transformers require significant memory.

Solution

Used smaller embedding dimensions and reduced batch size.

Challenge 4: Positional Encoding Confusion

Problem

Understanding sinusoidal encoding formulas.

Solution

Derived equations manually and tested outputs experimentally.

Challenge 5: Overfitting

Problem

Model memorized training data.

Solution

Applied dropout and reduced model complexity.

17. Experimental Observations

Observations During Training

1. Loss decreases steadily with epochs.
2. Attention improves contextual understanding.
3. Multi-head attention captures multiple relationships.
4. Residual connections stabilize training.
5. Layer normalization improves convergence.

Key Findings

- Transformers learn contextual patterns effectively.
 - Abstraction simplifies experimentation.
 - Modular code improves scalability.
-

18. Advantages and Limitations of Transformers

Advantages

1. Parallel processing
2. Better long-range dependency handling
3. Faster training on GPUs
4. State-of-the-art NLP performance
5. Scalable architecture

Limitations

1. High computational requirements

2. Large memory consumption
 3. Requires massive datasets for large-scale performance
 4. Training cost can be expensive
-

19. Real World Applications

Transformers are used in:

Domain	Application
NLP	Chatbots, Translation
Healthcare	Medical report analysis
Finance	Fraud detection
Education	Intelligent tutoring
Cybersecurity	Threat analysis
Vision	Image classification
Robotics	Autonomous systems

20. Future Improvements

Future enhancements may include:

1. Adding Decoder architecture fully
 2. Using pretrained embeddings
 3. Training on larger datasets
 4. Implementing Beam Search
 5. Using GPU acceleration
 6. Integrating HuggingFace Transformers
 7. Adding masked language modeling
 8. Fine-tuning pretrained models
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21. Conclusion

This project focused on building a Transformer model using abstracted modular functions.

The implementation demonstrated:

- Understanding of Transformer architecture
- Attention mechanisms
- Multi-head attention
- Encoder structure
- Positional encoding
- Deep learning abstraction principles

The use of abstracted functions significantly improved:

- Code readability
- Maintainability
- Scalability
- Reusability

Through this project, important concepts in modern AI architectures were explored both theoretically and practically.

The Transformer architecture represents one of the most revolutionary advancements in Artificial Intelligence and continues to power modern large language models and advanced AI systems.
